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Intracellular Expression of a Fluorogenic DNA Aptamer Using Retron Eco2

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eLife Assessment

This study presents a method for expressing single-stranded DNA fluorescent aptamers in *E. coli* using a retron-based strategy. The evidence supporting the successful expression and folding of DNA aptamers is **solid**, with clear demonstration of fluorescence after extraction, though the aptamers do not function in living cells. The method represents an **important** technical advance that will likely become standard for DNA aptamer expression in bacterial systems.

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Abstract

DNA aptamers are short, single-stranded DNA molecules that bind specifically to a range of targets such as proteins, cells, and small molecules. Typically, they are utilized in the development of therapeutic agents, diagnostics, drug delivery systems, and biosensors. Although aptamers perform well in controlled extracellular environments, their intracellular use has been less explored due to challenges of expressing them in vivo. In this study, we employed the bacterial retron system Eco2, to express a DNA light-up aptamer in *Escherichia coli*. Our data confirms that structure-guided insertion of the aptamer domain into the non-coding region of the retron enables reverse transcription and biosynthesis of functional aptamer constructs in bacteria. The purified DNA aptamer synthesized under intracellular conditions shows comparable activity to a chemically synthesized control. Our findings demonstrate that retrons can be used to express short DNA aptamers within living cells, potentially broadening and optimizing their application in intracellular settings.

Introduction

Aptamers are single stranded nucleic acids that fold into distinct structures, allowing them to bind specifically to target ligands.¹ The ligand-binding sites of aptamers are tailored to the shape and charge of the ligand, enabling precise interactions. In particular, aptamers isolated through in vitro selection protocols serve as affinity reagents, positioning them as nucleic acid analogues of antibodies.^{2,3} As such, they offer distinct advantages over their protein-based counterparts such as small size, as well as rapid and cost-effective manufacturing.³ An area of particular interest is the development of fluorescent light-up aptamers (FLAPs),^{4–6} which specifically bind conditionally fluorescent dyes and activate their fluorescence.⁷ The advent of FLAPs has also significantly advanced the study of folding of aptamers, as efficient folding can be determined as a function of fluorescence output. Pioneering work by Jaffery and coworkers has led to proliferation of numerous FLAPs. Notable examples include the RNA-FLAPs Spinach,⁸ Broccoli,⁹ Corn,¹⁰ Mango,¹¹ Squash,¹² and Pepper.⁷ These fluorogenic RNA aptamers serve as RNA mimics of fluorescent proteins and have transformed cellular imaging techniques to locate and understand RNA dynamics in vivo.

In contrast to RNA aptamers, DNA aptamers offer enhanced stability, greater ease of synthesis and chemical modification, as well as reduced cost, thereby facilitating more straightforward production and broader applicability in diverse fields, including therapeutics and diagnostics.^{3,13} For example Lettuce, a DNA FLAP that mimics the fluorescent properties of GFP, holds great promise for pathogen detection or as intracellular sensors for tumour cell recognition.^{14–16} However, since most DNA aptamer studies focus on their development and characterization in vitro, their potential for applications inside living cells remains largely unexplored. A particular challenge in this context is the difficulty of biosynthesising DNA aptamers in cells. Indeed, the ability to transcribe single-stranded RNA from DNA-based vectors has decisively contributed to the development of diverse RNA FLAPs for in-cell applications, such as the Broccoli aptamer.⁹ Therefore, methods for the intracellular synthesis and functional studies of single-stranded DNA (ssDNA) aptamers are highly desirable. Only a few approaches for intracellular ssDNA synthesis have been reported, in all cases using exogenous systems such as phagemid vectors,¹⁷ phages,¹⁸ and eukaryotic retroviral reverse transcriptases.^{19–23}

A more integrated, efficient, and potentially safer alternative to exogenous ssDNA expression systems could be the use of endogenous bacterial platforms. Retrons are prokaryotic genetic elements that are part of the bacterial phage defence machinery.^{24–27} Each retron typically consists of three components, namely a ncRNA transcript, an RT, and an effector protein. The ncRNA transcript adopts a specific secondary structure, nested between two repeat regions at its 5' and 3' ends that are complementary and hybridize (Figure 1 [↗](#)). The RT recognizes the folded ncRNA and initiates reverse transcription from the 2'OH of a conserved guanosine, using the base-paired region as primer.²⁸ The reverse transcription proceeds until about halfway along the ncRNA. Simultaneously, the RNA template is degraded by RNase H1, except for a stretch of approximately 5–10 RNA bases, which remain hybridized to the cDNA.²⁴ This results in a hybrid RNA-DNA structure (RT-DNA), with the non-coding RNA and DNA elements *msr* and *msd*, respectively. *msr* and *msd* are covalently attached by a 2'-5' linkage at the conserved guanosine and the base paired region close to the 5' end of *msd* (3' end of *msr*). This ability of retons to synthesize abundant ssDNA in vivo has generated considerable interest in using them as an alternative to exogenously delivered DNA oligonucleotides in genome engineering and genome editing.^{28–35}

While retons have also been proposed as a potential method for expressing DNA aptamers,^{20,22} this approach has not yet been experimentally validated. Recent studies have indicated that retons can indeed be leveraged to express functional ssDNA in vivo. For example, Lopez et al tested modifications to the ncRNA in case of retron Eco1 in order to boost RT-DNA abundance.³² More recently, Liu et al also used retron Eco1 to express 10–23 DNAzyme in vivo.³⁶ However, it is poorly understood how the encoding of a functional ssDNA sequence in retons will affect its activity. Such studies may lead to valuable design principles that would be especially important in the cellular context, as the RT-DNA is known to interact with not only the RT, but, depending on the retron, also RNaseH and the retron associated effector proteins.^{24,25}

In this work we sought to demonstrate expression of a DNA FLAP Lettuce as a model using retron Eco2.³⁷ We first characterize expression of plasmid-borne wild type Eco2, determining abundance and intracellular stability of Eco2 RT-DNA. Subsequently, we test suitable integration sites of different length variants of Lettuce within the retron *msd* using in vitro prototyping. After identifying effective insertion positions, we finally demonstrate fluorogenic activity of the expressed RT-DNA, embedded within the Lettuce aptamer. Our work demonstrates proof of concept of expressing DNA aptamers using retons.

Results

We first established an expression system for retron Eco2 by cloning the ncRNA and RT in a pET28a plasmid backbone, under the control of a T7 promoter. The plasmid was transformed in BL21AI cells, that enable tight control of T7-based expression. The intracellular synthesis of Eco2 RT-DNA expression was induced with 1 mM IPTG and 0.2% arabinose (Methods) and confirmed by denaturing PAGE of DNA extracts from induced cells. Using gel-based quantification of both RT-

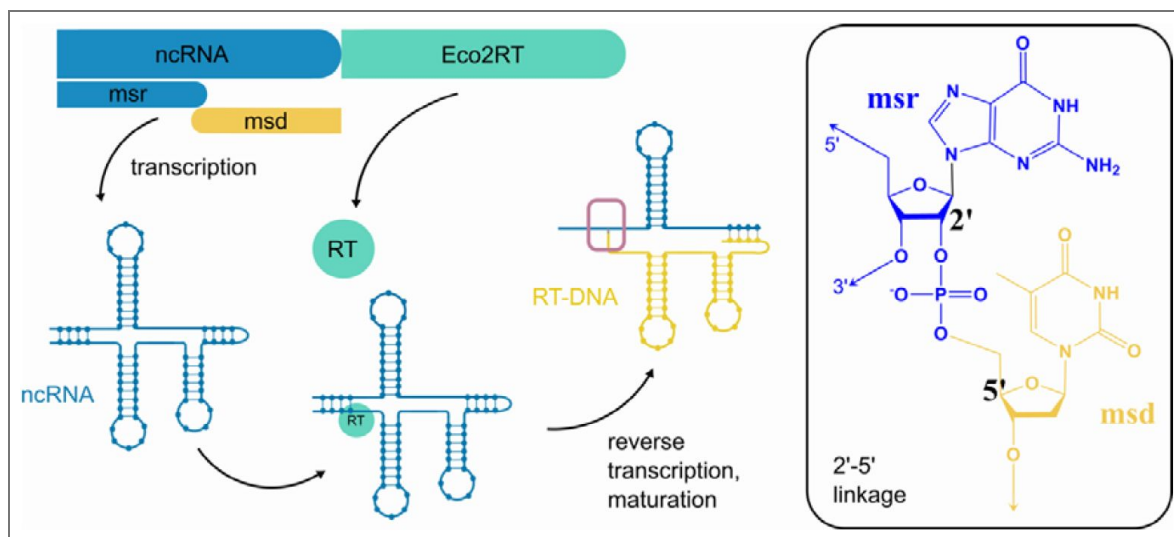


Figure 1. Schematic of the *Eco2* gene locus, which encodes the non-coding RNA (*ncRNA*, blue) and the *Eco2* reverse transcriptase gene (*Eco2RT*, green).

The *msd* (yellow) region of the *ncRNA* is reverse transcribed into *msd* DNA, which is linked to the remaining *msr* via a 2'-5' linkage conserved guanosine close to the 3'-end of the *msr*, as indicated in the inset.

DNAs, we observed that Eco2 RT-DNA expression levels were approximately 3-4 times higher than those of genomically encoded Eco1 RT-DNA. To quantify expression levels, we further characterized Eco2 RT-DNA expression using qPCR. Upon induction with IPTG and arabinose, we observed that the copy-number ratio between amplicon 1 (which can arise from amplification of both single-stranded RT-DNA and plasmid DNA, as shown in Figure 2A) and amplicon 2 (which can only be amplified from the plasmid construct) increased by almost eightfold in induced compared to uninduced cells. Assuming an average copy number of 15-20 for the pET28b expression, which utilizes a pBR322 origin³⁸, we estimate that 100-200 single-stranded RT-DNA molecules are present in induced cells. With an average cytoplasmic volume of 7×10^{-16} L of an *E. coli* cell³⁹, this corresponds to an intracellular concentration of approximately 250 – 500 nM.

Because no data are available on the stability of single-stranded RT-DNA in cells, we next determined the half-life of Eco2 RT-DNA (Supplementary Information). To this end, induced cells were thoroughly washed to remove residual IPTG and arabinose, thereby preventing de novo expression of RT-DNA. The washed cells were then resuspended in fresh medium and allowed to continue growing. RT-DNA levels were quantified by qPCR at defined time points. The gradual decrease in RT-DNA abundance could thus be attributed to either active degradation (e.g. by intracellular nucleases) or dilution due to cell growth and division. Intriguingly, no evidence for active RT-DNA degradation was observed as the decrease in RT-DNA could be entirely explained by dilution (Figure 2–figure supplement 1) indicating exceptional intracellular stability.

Having established the inducible expression of plasmid-encoded Eco2 in *E. coli*, we next sought to explore its suitability as a scaffold to host a functional Lettuce aptamer. Expression of ssDNA using retrons is typically achieved by encoding a cargo gene in the *msd* region of the ncRNA. This approach has been successfully demonstrated for recombineering donors³², protein-binding DNA sequences⁴⁰ and DNazymes³⁶. However, it was not clear as to what extent the folding of the cargo aptamer sequence would be affected due to the extensive secondary structure of the Eco2 *msr-msd* RNA-DNA hybrid.

We speculated that the position of the Lettuce aptamer domain in the *msd* region would significantly affect its ability to fold into the functional three-dimensional structure. To test this hypothesis, we used a structure-guided approach to insert different variants of the Lettuce aptamer as cargo into four different positions in the *msd* region of retron Eco2 (Figure 3). To identify four suitable insertion sites in the *msd*, we used the minimal free energy structure predicted for the Eco2 wild type *msd* by RNAfold⁴¹ with energy parameters for DNA as guide to identify single stranded and / or loop regions where an insertion of the Lettuce domain was expected to not interfere with the native fold of the remaining *msd* sequence. We omitted the *msr*-overlapping region in the input sequences as it is known to form a duplex with the RT-DNA as well as the *msr* RNA, as to the best of our knowledge, there is no algorithm capable of simulating mixed RNA-DNA sequences that are connected via a 2'-5' bond. The positions v1 and v3 are in two different predicted loop regions. The insertion site v2 is located in a single stranded that connects two hairpin loops, whereas position v4 directly upstream of the predicted base-paired region between the *msd* and *msr* sequences.

The P1 stem of the original lettuce aptamer can be truncated down to four base pairs without significantly affecting its fluorescence output upon DFHBI-1T binding.¹⁴ We made use of this property in order to minimize the impact of Lettuce insertion on RT-DNA maturation while maintaining the ligand binding activity of Lettuce. Initially, we probed if truncated aptamer variants with P1 stem-lengths of 4, 8, and 11 base pairs (designated 4L, 8L, and 11L) along with the full-length aptamer (FL) can still fold after embedding them into the integration sites of the emulated Eco2 RT-DNA.

Using chemically synthesized RT-DNA-Lettuce fusion constructs as model constructs, we first tested the fluorogenic potential of 4L after its insertion into the above-mentioned positions v1 to v4 of the Eco2 RT-DNA (resulting in the constructs 4LEv1 – 4LEv4) using in gel staining with DFHBI-1T (Figure 3–figure supplement 1). We observed only very low levels of in-gel fluorescence compared to a standard full-length Lettuce (FL) or 4L alone, indicating that 4L folding was hampered upon embedding in the retron *msd* scaffold. In contrast, we observed a pronounced

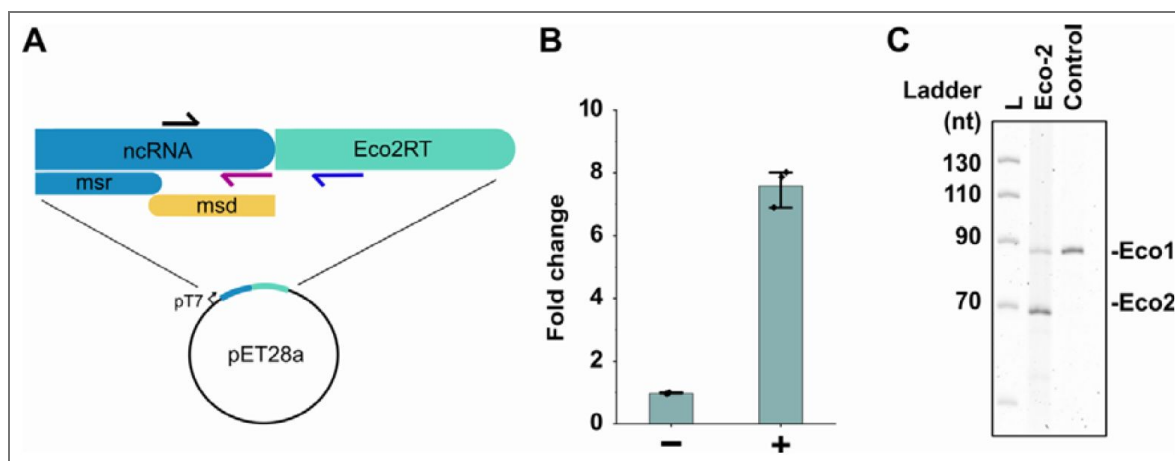


Figure 2.

(A) Illustration of primer target sites for qRT-PCE experiments to determine RT-DNA abundance. The black arrows indicate the forward primer that pairs either with the purple reverse primer (amplify both RT-DNA and plasmid DNA) or the blue reverse primer that only amplified the plasmid DNA. (B) Fold enrichment of the RT-DNA/plasmid template over the plasmid alone upon induction, as measured by qPCR; Unpaired t-test, induced versus uninduced: $p < 0.0001$; $n = 3$ biological replicates. (C) A TBE-Urea polyacrylamide gel, stained with SYBR Gold showing RT-DNA corresponding to retron Eco1 (90 nt) and retron Eco2 (70 nt).

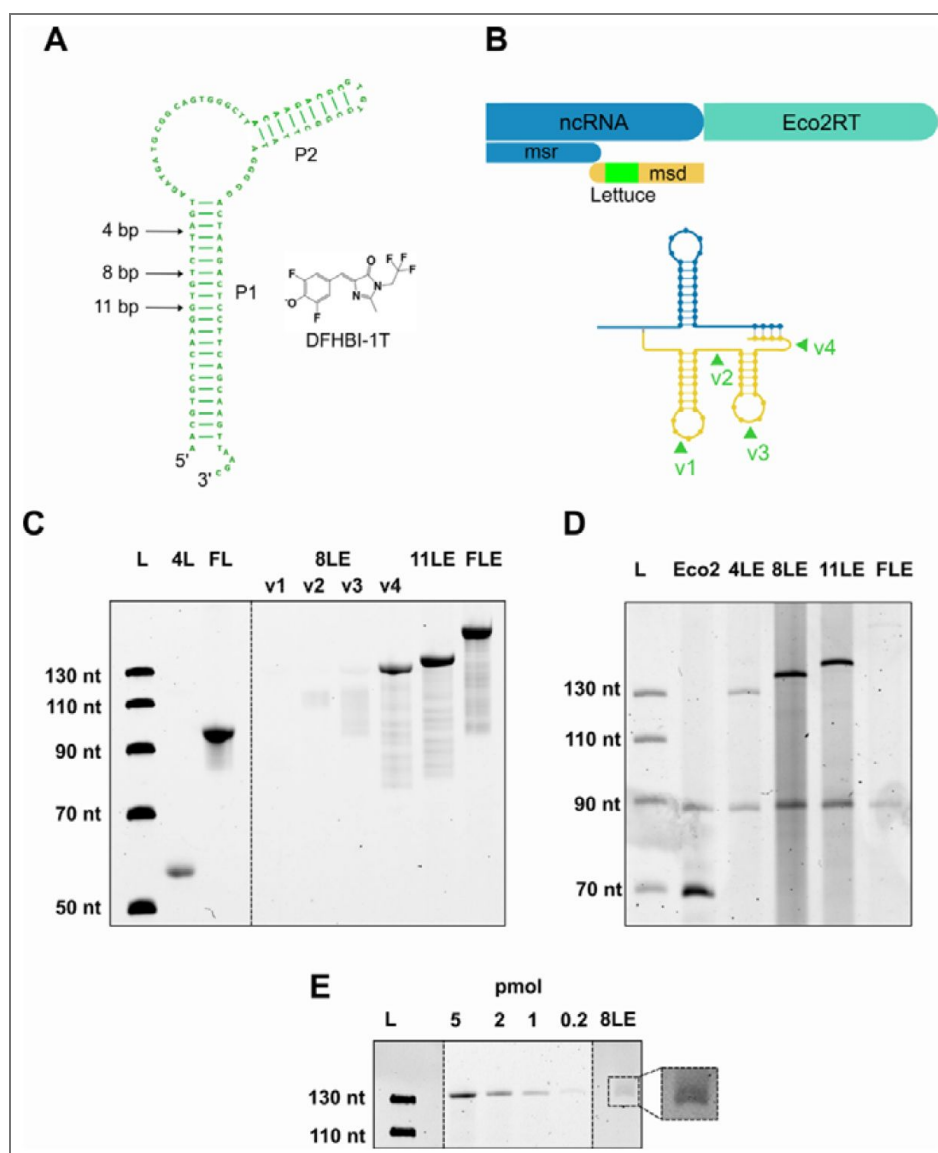


Figure 3.

(A) Schematic of Lettuce aptamer. The arrows indicate the length to which the P1 stem can be shortened without significant loss of FLAP functionality. (B) Schematic of insertion of Lettuce aptamer sequence in 4 distinct positions in the *msd* region of retron *eco2*. The ssDNA structure was simulated using Vienna RNA fold with DNA parameters. Lettuce aptamer sequence was inserted at single stranded and/or loop regions to minimise interference with the native fold of the *msd* sequence. (C) DFHBI-1T stained TBE urea-PAGE showing fluorescence of different oligonucleotides mimicking the Lettuce-Eco2 (LE) fusion constructs with varying P1 lengths: 8LE v1-v4 (134 nt), 11LE (140 nt), and FLE (166 nt). Free Lettuce with either the 4 nt P1 stem (4L) or full-length P1 (FL) served as positive controls. (D) SYBR stained TBE urea-PAGE of extracted Eco2 wild type and Eco2-Lettuce RT-DNA fusions after expression in *E. coli*. In all variants, the lettuce aptamer was inserted into position v4 of the *msd* scaffold. (E) DFHBI-1T stained TBE urea-PAGE showing fluorescence of 8LEv4 RT-DNA purified from *E. coli* cells. Lanes with different loading amounts in pmol of the synthetic RT-DNA standards are shown as comparison.

fluorescence signal for 8L when it was inserted position v4 (Figure 3C [↗](#)), demonstrating that this insertion site was compatible with Lettuce folding given a sufficient length of P1. Similarly, we observed strong in-gel fluorescence signals for 11LEv4 and FLEv4 comparable with native full-length Lettuce.

To cross-validate these results, we determined the dissociation constants between DFHBI-1T and the chemically synthesized Eco2-Lettuce v4 surrogate constructs using the reported Lettuce binding buffer (40 mM HEPES pH 7.5, 100 mM KCl, and 1 mM MgCl₂)¹⁴. When increasing concentrations of each oligonucleotide were incubated in binding buffer containing a constant concentration of DFHBI-1T, we obtained well-defined fluorescence responses for all v4 constructs (Figure 3–figure supplement 2). When fitting the binding isotherms with a quadratic association curve (Supplementary Methods) we obtained dissociation constants (K_d) for 8LE, 11LE, FLE that were comparable to the unembedded full-length aptamer (FL) ~ 0.12 μ M (0.01, 2.46, 95% CI). Binding was weaker for the synthesized ssDNA mimic 4LE-v4 with a less well-defined K_d of ~ 6 μ M (4.92, 7.18, 95% CI), which was well in agreement with our in-gel fluorescence experiments.

Having confirmed that the v4 surrogate of 8LE, 11LE and FLE constructs maintained the light-up aptamer properties of the embedded Lettuce sequences, we investigated whether the different length variants could be expressed in *E. coli* after integration into the plasmid-borne *msd* region of retron Eco2. PAGE analysis of RT-DNA from induced *E. coli* BL21-AI expressing the Eco2-Lettuce fusion constructs confirmed complete synthesis of the 8LE (134 nt) and 11LE (140 nt) species, and low-level production of FLE (166 nt), despite their *msd*-encoded RT-DNAs being more than twice as long as the wild-type (67 nt). Overall, the insertion of the Lettuce aptamer domain into the *msd* region of retron Eco2 resulted in a decrease in RT-DNA level compared to wild-type Eco2: While the length-corrected intensity of wild-type Eco2 was 3-fold higher than that of the endogenous Eco1 band, the 4LE, 8LE, 11LE and FLE constructs showed between $\sim 25\%$ (FLE) and ~ 65 –90% (4LE, 8LE, and 11LE) of the length-corrected relative band intensities (Figure 3 [↗](#)).

Next, we tested whether heterologously expressed 8LEv4 RT-DNA retains its fluorogenic properties, as it remained unknown whether retron-derived aptamer-DNAs produced under intracellular conditions are functionally equivalent to chemically synthesized ssDNA oligonucleotides. Because column-based purification could not be scaled efficiently due to the large amounts of plasmid DNA suppressing column binding of the short ssDNAs, we developed a modified TRIzol-based extraction protocol optimized for ssDNA isolation while minimizing co-purification of plasmid DNA. Although the new method yielded considerably cleaner RT-DNA preparations, overall recovery was lower, necessitating the use of larger culture volumes to obtain sufficient material for analysis. The new extraction method was evaluated in pilot experiments for the 8LEv1-4 variants, and PAGE analysis of the isolated RT-DNA from pilot-scale experiments confirmed full-length synthesis of all four position variants (Figure 3–figure supplement 3). This approach was then applied to larger volumes for preparative extractions. TBE-urea PAGE analysis with DFHBI-1T staining indeed revealed that the purified, intracellular ssDNA retains fluorogenic activity upon DFHBI-1T binding.

While no measurable fluorescence was detected in induced cell cultures expressing 8LEv4 - aside from DFHBI-1T-independent expression-related autofluorescence (Figure 3–figure supplement 4), the intracellular production of a fluorogen-binding RT-DNA demonstrates that retron-encoded ssDNA can, in principle, be used to generate functional aptamers in a cellular environment. The limited in vivo signal likely reflects both the moderate RT-DNA abundance (~ 100 –200 molecules per cell) and the substantially lower fluorogenic efficiency of Lettuce compared with the RNA light-up aptamer Broccoli (~ 100 -fold, Figure 3–figure supplement 5), which is optimized for in vivo imaging of RNA⁹. Nevertheless, the detection of fluorogenic activity in purified material establishes a proof-of-principle that retron-derived ssDNA can be used to display DNA aptamers inside living cells.

Discussion

Bacterial retrons such as Eco2 provide a versatile platform for intracellular reverse transcription-based synthesis of single-stranded DNA. To our knowledge, our present study is the first to show that functional DNA aptamers that recognize small molecule ligands can also be expressed via retrons. Expanding the repertoire of retrons toward DNA aptamer expression promises to significantly expand the repertoire of functional non-coding DNAs and enable the creation of new sensors and regulators with a wide range of intracellular applications. At the same time, retron platforms provide a direct way to tailor DNA aptamers specifically to intracellular conditions. This is particularly important as our data show that both the low expression levels in combination with the moderate fluorogenic activity of Lettuce provide ample room for, e.g. evolutionary improvement of retron-encoded DNA aptamer performance under intracellular conditions. We observed a significant decrease in Eco2 expression levels after integration of the Lettuce domain, suggesting that reverse transcription is hindered by the changes in the *msd* sequence.³²

In Eco2, the RT is directly fused to a topoisomerase-primase-like (Toprim) effector domain, whose activity causes abortive infection.^{42,43} Recent structural studies elucidated the structure of the Eco2 nucleoprotein complex and revealed that each RT-Toprim fusion protein forms a 1:1 complex with the 2'-5' linked DNA-RNA hybrid that further assembles into a trimer.⁴² Interestingly, while the *msr* portion of each complex bridges adjacent RT-Toprim monomers and is integral to the structural integrity of the complex, the electron density of the *msd* beyond the stem downstream of the branched 2'-5' linkage remained largely undefined, suggesting a strong degree of flexibility. The proximal *msd* stem is proposed to sterically block the nuclease active site of the Toprim effector domain; upon activation by the phage-encoded ssDNA nuclease DenB from T4-related phages, this inhibition is relieved, enabling Toprim to cleave tRNA and thereby inducing translational shutdown.^{42,43} The electron density of the distal *msd* region was largely unresolved in the reported cryo-EM structure, indicating that it is not an integral structural element of Eco2 and likely tolerates mutations or insertions without preventing trimerization. In our study, we indeed observed no major growth defects upon insertion of the aptamer domain at positions v1 to v4 confirming the high degree of tolerance of the *msd* region distal to the 2'-5' branch point. This suggests that alternative functional sequences could be similarly integrated into the distal *msd* segment, provided that flanking *msr* and *msd* regions do not disrupt folding. A major advantage of retron-based intracellular synthesis of ssDNA in vivo is its simplicity. The desired sequence can be cloned into the *msd* region of plasmid-borne or genomically encoded retron Eco2. The copy number of the aptamer can also, in principle, be tuned by inducer concentration, promoter strength and plasmid copy number. An ever-increasing number of DNA aptamers have been generated against a wide range of targets, from small molecules, macromolecules to whole cells.^{3,44} The retron system used in this work could enable the exploration of cellular applications for this growing repertoire of DNA aptamers. A further advantage of expressing DNA aptamers with retrons is the exceptional intracellular stability of the *msd*-derived ssDNA in the retron complex (Supplementary information, Figure 2—figure supplement 1). Finally, expression of related retron systems has been demonstrated in yeast⁴⁵ and mammalian cells^{32,46}, thus potentially widening the range of in vivo applications of DNA aptamers.

In conclusion, retron-based aptamer expression platforms present a promising avenue for the development of novel DNA-based applications. Future research aimed at optimizing intracellular synthesis may elucidate the conditions under which larger single-stranded DNAs can be synthesized at higher concentrations without compromising reverse transcription and may also identify the key factors influencing the optimal in vivo folding of cargo sequences.

Methods

Constructs and strains

The retron Eco2 sequence (*msr-msd* and RT) was obtained from Millman et al.²⁶, and purchased from IDT as gblock, which was inserted into a pET28a vector under control of T7 promoter and lac operator. Plasmid assembly was performed using the HiFi DNA Assembly (New England Biolabs) according to the supplier's instructions. The resulting plasmids were transformed into chemically competent *E. coli* Top 10 cells. Plasmid sequences were verified by colony PCR and Sanger sequencing (Microsynth AG). Verified plasmids were then transformed into chemically competent BL21AI cells for further experiments. These cells harbor endogenous retron Eco1 which serves as an internal control and a cassette encoding T7 polymerase driven by arabinose-inducible araBAD promoter. Lettuce variant sequences were cloned into retron Eco2 plasmid using Q5® Site-Directed Mutagenesis Kit (New England Biolabs). Primer sequences and annealing temperatures were generated using the NEBaseChanger™ tool.

RT-DNA extraction

Miniprep method

5 ml cultures of BL21AI cells with corresponding Lettuce constructs (LB with selection antibiotic) were inoculated with cells from a single colony and grown at 37°C, 220 rpm for 90 min. They were then induced with 1 mM IPTG and 0.2% arabinose, followed by overnight expression. OD₆₀₀ of the overnight culture was measured and cells corresponding to 2 ml of OD₆₀₀ = 1 were used for miniprep. A NEB Monarch® Plasmid Miniprep Kit was used for RT-DNA extraction (New England Biolabs). Following miniprep, cells were eluted in 40 µl milliQ purified water. The miniprep product was then treated with a RNase Cocktail™ Enzyme Mix (Invitrogen) at 37°C for 30 min. The RNase treated samples were then purified using a ssDNA/RNA Clean & Concentrator kit (Zymo).

The purified RT-DNA was analysed on a 10% TBE-Urea polyacrylamide gel with a 1x TBE running buffer. Gels were stained with SYBR Gold (Invitrogen) and imaged on Sapphire Biomolecular Imager (Azure biosystems). Band intensities were normalized by length of RT-DNA. RT-DNA abundance of Eco2 and Lettuce variants was quantified by determining fold-change compared to the Eco1 band. Band intensity analysis was performed with ImageJ⁴⁷.

TRIzol based extraction method

25 mL pre-cultures of BL21A cells transformed with the pET28a plasmid carrying the corresponding Lettuce variant were seeded from a single colony and grown in LB-medium containing 50 µg/mL kanamycin at 37 °C at 220 rpm for four hours until to an of OD₆₀₀ of 0.5-0.8. These were used to inoculate 750 mL cultures, which were then grown at 37°C and 150 rpm for two hours. The cultures were then induced with IPTG (1 mM) and arabinose (0.2%) and allowed to grow overnight (~16 h) until they reached an OD₆₀₀ 2.4 - 2.9 and harvested by centrifugation at 4°C and 6000 g for 30 min. The cell pellets were then resuspended in TES buffer and lysozyme (Ready-Lyse Lysozyme Solution, Biozym) was added according to the manufacturer's instructions (25 ml TES + 4.5 µL 30,000 U/µL Lysozyme). The samples were then incubated for 15 minutes at room temperature before they were split into 12 x 50 mL falcon tubes. 20 mL of TRIzol was added to each tube and the solutions were gently vortexed until they became clear. Next, 4 mL chloroform was added and the samples were vortexed until the solutions were homogeneous. The samples were then incubated at room temperature for 5 min until phase separation was observed. Subsequently, the samples were centrifuged for 30 min at 4°C and 12,000 g, after which a volume of 10 ml of the supernatant was added to 10 ml isopropanol. The samples were vortexed, then incubated for two hours at -20 °C. They were then centrifuged at 30,000 g and 4°C for 1 hour. The supernatant was then removed, and the pellet was washed twice with 10 ml of 70% ethanol and centrifuged at 4 °C for 5 minutes at 10,000 g. The pellets were air-dried for 15 minutes and then resuspended in 1000 µL of ddH₂O.

In-gel staining of the RT-DNA

After isolation using the above-described TRIzol-based method, 650 μL of the total RNA / retron (1000 μL) material was incubated with 250 mM NaOH and 6.25 mM EDTA at 80 $^{\circ}\text{C}$ for 1 hour. After incubation, the pH was neutralised by the addition of 250 mM HCl, after which the ssDNA was purified using the ssDNA/RNA Clean & Concentrator Kit (Zymo Research) following the manufacturer's instructions. A total of three columns were used to isolate these samples, with 9 μL of ddH₂O for elution. The eluate from column 1 was used to elute DNA from columns 2 and 3 to enhance the total RT-DNA concentration. The entire sample (approximately 7.5 μL), was mixed with 15 μL of loading dye (10 mM EDTA in formamide) and heated to 95 $^{\circ}\text{C}$ for three minutes, followed by immediate incubation on ice for five minutes. The sample was then loaded into a single well of a 10% TBE urea-PAGE gel and run for 80 minutes at a constant power of 20 W. The gel was then washed twice with ddH₂O to remove the urea and EDTA. This was followed by a 30-minute stain with 50 mL DFHBI-1T staining solution, containing 40 mM HEPES pH 7.5, 100 mM KCl, 1 mM MgCl₂ and 10 μM DFHBI-1T. After a final five-minute wash in ddH₂O to remove excess dye, the gel was imaged at 488/518 nm (DFHBI-1T) and 520/565 nm (Cy3-tagged ladder) on a Sapphire Bioimager.

qPCR

qPCR experiments were performed using Techne Prime Pro 48 Real-Time qPCR system, according to comparative C_T method,^{48,49} as described by Lopez et al.³² For qPCR analysis, cell cultures grown for ~16h were used. 25 μL of bacterial culture at an OD₆₀₀=1 was mixed with 25 μL water and incubated at 95 $^{\circ}\text{C}$, for 5 min to lyse cells. 0.3 μL of this boiled culture was used as templates in 30 μL qPCR reactions using Luna® Universal qPCR Master Mix (New England Biolabs). qPCR quantification of RT-DNA abundance was performed using a set of three primers. Two primers were complementary to the *msd* region of Retron Eco2 and the third was an outnest primer complementary to non-retron elements of the expression plasmid. Two templates were thus amplified: One corresponding to an RT-DNA sequence that could use both the RT-DNA and the plasmid as templates, and a second that could use only the plasmid-encoded Eco2 as a template. The difference between cycle threshold (ΔC_T) of the inside (RT-DNA) and outside (plasmid) primer sets was then calculated. This ΔC_T was then subtracted from the ΔC_T of uninduced samples. Fold enrichment of RT-DNA/plasmid over plasmid alone was then calculated as $2^{-\Delta\Delta\text{C}_T}$, for each biological replicate. Presence of RT-DNA results in a fold change > 1.

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Additional information

Author Contributions

M.V., C.M., D.S., H.M. designed research; M.V., C.M., S.K., M.D. performed research; M.V., C.M., K.B., H.M., analysed data; K.B. contributed new reagents/analytic tools; M.V., C.M. and H.M wrote the paper.

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Additional files

[Supplementary Information](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors use an interesting expression system called a retron to express single-stranded DNA aptamers. Expressing DNA as a single-stranded sequence is very hard - DNA is naturally double stranded. However, the successful demonstration by the authors of expressing Lettuce, which is a fluorogenic DNA aptamer, allowed visual demonstration of both expression and folding, but only after extraction in cells, but not in vivo (possibly because of the low fluorescence of Lettuce, or perhaps more likely, some factor in cells preventing Lettuce fluorescence). This method will likely be the main method for expressing and testing DNA aptamers of all kinds, including fluorogenic aptamers like Lettuce and the future variants / alternatives.

Strengths:

This has an overall simplicity which will lead to ready adoption. I am very excited about this work. People will be able to express other fluorogenic aptamers or DNA aptamers tagged with Lettuce with this system.

Weaknesses:

Some things could be addressed/shown in more detail, e.g. half-lives of different types of DNA aptamers and ways to extend this to mammalian cells.

<https://doi.org/10.7554/eLife.99554.2.sa2>

Reviewer #2 (Public review):

Summary:

This manuscript explores a DNA fluorescent light up aptamer (FLAP) with the specific goal of comparing activity in vitro to that in bacterial cells. In order to achieve expression in bacteria, the authors devise an expression strategy based on retrons and test four different constructs with the aptamer inserted at different points in the retron scaffold.

The initial version of this manuscript made several claims about the fluorescence activity of the aptamers in cells, and the observed fluorescence signal has now been found to result from cellular auto-fluorescence. Thus, all data regarding the function of the aptamers in cells have been removed.

Negative data are important to the field, especially when it comes to research tools that may not work as many people think that they will. Thus, there would have been an opportunity here for the authors to dig into why the aptamers don't seem to work in cells.

In the absence of insight into the negative result, the manuscript is now essentially a method for producing aptamers in cells. If this is the main thrust, then it would be beneficial for the authors to clearly outline why this is superior to other approaches for synthesizing aptamers.

<https://doi.org/10.7554/eLife.99554.2.sa1>

Author response:

The following is the authors' response to the original reviews

Comment to both reviewers:

We are very grateful for the thoughtful and constructive comments from both reviewers. During the revision, and in direct response to these comments, we performed additional control experiments for the cellular fluorescence measurements. These new data revealed that the weak increase in green fluorescence reported in our original submission does not depend on retron-expressed Lettuce RT-DNA or the DFHBI-1T fluorophore, but instead reflects stress-induced autofluorescence of *E. coli* (e.g. upon inducer and antibiotic treatment).

We also benchmarked the fluorogenic properties of Lettuce against the RNA FLAP Broccoli and found that Lettuce is ~100-fold less fluorogenic under optimal in vitro conditions. Consequently, with the currently available, in vitro- but not in vivo-optimized Lettuce variants, intracellular fluorescence cannot be reliably detected by microscopy or flow cytometry. We have therefore removed the original flow cytometry / and in-culture-fluorescence data and no longer claim detectable intracellular Lettuce fluorescence.

In the revised manuscript, we now directly demonstrate that retron-produced Lettuce RT-DNA can be purified from cells and remains functional ex vivo with a gel-based fluorophore-binding assays. Together, these data clarify the current limitations of DNA-based FLAPs for in vivo imaging, while still establishing retrons as a viable platform for intracellular production of functional DNA aptamers.

Reviewer #1 (Public Review):

Summary:

The authors use an interesting expression system called a retron to express single-stranded DNA aptamers. Expressing DNA as a single-stranded sequence is very hard - DNA is naturally double-stranded. However, the successful demonstration by the authors of expressing Lettuce, which is a fluorogenic DNA aptamer, allowed visual demonstration of both expression and folding. This method will likely be the main method for expressing and testing DNA aptamers of all kinds, including fluorogenic aptamers like Lettuce and future variants/alternatives.

Strengths:

This has an overall simplicity which will lead to ready adoption. I am very excited about this work. People will be able to express other fluorogenic aptamers or DNA aptamers tagged with Lettuce with this system.

We thank the reviewer for their thoughtful assessment and appreciate their encouraging remarks.

Weaknesses:

Several things are not addressed/shown:

(1) How stable are these DNA in cells? Half-life?

We thank the reviewer for this insightful question.

Retron RT-DNA forms a phage surveillance complex with the associated RT and effector protein[1-4]. Moreover, considering the unique ‘closed’ structure of RT-DNA[5] (with the ends of *msr* and *msd* bound either by 2’-5’ linkage and base paired region) and its noncoding function, we hypothesized that the RT-DNA must be exceptionally stable. Nevertheless, we attempted to determine half-life of the RT-DNA using qPCR for Eco2 RT-DNA. To this end, we designed an assay where we would first induce RT-DNA expression, use the induced cells to start a fresh culture without the inducers. We would then take aliquots from this fresh culture at different timepoints and determine RT-DNA abundance by qPCR.

We induced RT-DNA expression of retron Eco2 in BL21AI cells as described in the Methods. After overnight induction, cells were washed to remove IPTG and arabinose, diluted to $OD_{600} = 0.2$ into fresh LB without inducers, and grown at 37°C. At the indicated time points, aliquots corresponding to $OD_{600} = 0.1$ were boiled (95°C, 5 min), and 1 μ L of the lysate was used as template in 20 μ L qPCR reactions (see revised Methods for details).

Assuming RT-DNA degradation would occur by active degradation mechanisms (nuclease-mediated degradation) and dilution (cell growth and division), we determined the rate of degradation by the following equation

$$C(t) = 1 + A \cdot e^{-kt} \frac{OD_{600}(t_0)}{OD_{600}(t)} \quad (1)$$

where k is the degradation rate constant and the ratio $\frac{OD_{600}(t_0)}{OD_{600}(t)}$ is the dilution factor which takes into account dilution by cell division. $OD_{600}(t)$ was determined by fitting the OD_{600} measurements by the following the equation describing logistic growth:

$$OD_{600}(t) = \frac{OD_{max}}{1 + e^{-r \cdot (t - t_0)}} \quad (2)$$

Which yields the plots shown in Figure 2–figure supplement 1.

After substituting $OD_{600}(t)$ by the function in equation (2), we fit the experimental data for the fold-change of the RT-DNA to equation (1). Interestingly, the best fit (red) was obtained with a converging towards zero suggesting that the half-life of the RT-DNA is beyond the detection limit of our assay. To showcase typical half-lives of RNA, which are in the range of minutes in growing *E. coli* cells[6], we refitted the data using constant half-life of 15 and 30 minutes. In both cases, simulated curve deviated significantly from the experimental data further confirming that the half-life of the RT-DNA is probably orders of magnitude higher than the doubling time of *E. coli* under these optimal conditions. While we cannot exclude that the RT-DNA is still produced as a result of promotor leakiness, but we expect this effect to be low as the expression of RT-DNA in *E. coli* AI cells requires both the presence of IPTG and arabinose, which were thoroughly removed before inoculating the growth media with the starter culture. Overall, our data therefore argues for an exceptional stability of the RT-DNA in growing bacterial cells.

We have now included this new experimental data in the supplementary information.

(2) What concentration do they achieve in cells/copy numbers? This is important since it relates to the total fluorescence output and, if the aptamer is meant to bind a protein, it will reveal if the copy number is sufficient to stoichiometrically bind target proteins. Perhaps the gels could have standards with known amounts in order to get exact amounts of aptamer expression per cell?

The copy number of RT-DNA can be estimated based on the qPCR experiments. We use a pET28a plasmid, which is low-copy with typical copy number 15-20 per cell[7]. We determined the abundance of RT-DNA over plasmid/RT-DNA, upon induction, to be 8-fold, thereby indicating copy number of Eco2 RT-DNA to be roughly around 100-200. Assuming an average aqueous volume of *E. coli* of 1 femtoliter[6], the concentration of RT-DNA is ~250-500 nM. We have added this information to the revised version of the manuscript.

(3) Microscopic images of the fluorescent E. coli - why are these not shown (unless I missed them)? It would be good to see that cells are fluorescent rather than just showing flow sorting data.

In the original submission, we used flow cytometry as an orthogonal method to quantify the fluorescence output of intracellularly expressed Lettuce aptamer, anticipating that it would provide high-throughput, quantitative information on a large population of cells. During the revision, additional controls revealed that the weak increase in fluorescence we had previously attributed to Lettuce expression was in fact a stress-induced autofluorescence signal that occurred independently of retron RT-DNA and DFHBI-1T. We have therefore removed these data from the manuscript and no longer claim detectable intracellular Lettuce fluorescence.

To understand this limitation, we compared the in vitro fluorescence of Lettuce with that of the RNA FLAP Broccoli, which is commonly used for RNA live-cell imaging. Under optimal in vitro conditions, Lettuce shows ~100-fold lower fluorescence output than Broccoli (new Figure 3–figure supplement 5). Given this poor fluorogenicity and the low intracellular concentration of retron RT-DNA (now derived from the qPCR experiments), we conclude that the current Lettuce variants are below the detection threshold for in vivo imaging in our system. We now explicitly discuss this limitation and the need for further (in vivo) evolution of DNA-based FLAPs in the revised manuscript.

(4) I would appreciate a better Figure 1 to show all the intermediate steps in the RNA processing, the subsequent beginning of the RT step, and then the final production of the ssDNA. I did not understand all the processing steps that lead to the final product, and the role of the 2'OH.

We thank the referee for this comment. We have now made changes to Figure 1, showing the intermediate steps as well as a better illustration of the 2'-5' linkage.

(5) I would like a better understanding or a protocol for choosing insertion sites into MSD for other aptamers - people will need simple instructions.

We appreciate the reviewer for bringing up this important point. We simulated the ssDNA structure using Vienna RNA fold with DNA parameters. Based on the resulting structure, we inserted Lettuce sequence in the single stranded and/or loop regions to minimise interference with the native msd fold. We have now included this information in the description of Figure 3.

(6) Can the gels be stained with DFHBI/other dyes to see the Lettuce as has been done for fluorogenic RNAs?

Yes. We have now included experiments where we performed in-gel staining with DFHBI-1T for both chemically synthesized Eco2-Lettuce surrogates as well as the heterologously expressed Eco2-Lettuce RT-DNA. We have added this data to the revised Figure 3 (panel C and E).

(7) Sometimes FLAPs are called fluorogenic RNA aptamers - it might be good to mention both terms initially since some people use fluorogenic aptamer as their search term.

We thank the referee for this useful suggestion. We have now included both terms in the introduction of the revised version.

(8) What E coli strains are compatible with this retron system?

Experimental and bioinformatic analysis have shown that retrons abundance varies drastically across different strains of *E. coli*[8-10]. For example, in an experimental investigation of 113 independent clinical isolates of *E. coli*, only 7 strains contained RT-DNA[8]. In our experiments, we have found that BL21AI strain is compatible with plasmid-borne Eco2. The fact that this strain has a native retron system (Eco1) allowed us to use it as internal standard. However, we were also able express Eco2 RT-DNA in conventional lab strains such as *E. coli* Top 10 (data not shown), indicating both ncRNA and the RT alone are sufficient for intracellular RT-DNA synthesis.

(9) What steps would be needed to use in mammalian cells?

We appreciate the reviewer's thoughtful inquiry. Expression of retrons has been demonstrated in mammalian cells by Mirochnitchenko et al[11] and Lopez et al[12]. For example, Lopez et al demonstrate expression of retrons in mammalian cell lines using the Lipofectamine 3000 transfection protocol (Invitrogen) and a PiggyBac transposase system[12]. We also mention this in the discussion section of the revised manuscript. Expression of retron-encoded DNA aptamers in mammalian cells should be possible with these systems.

(10) Is the conjugated RNA stable and does it degrade to leave just the DNA aptamer?

We are grateful to the reviewer for their perceptive question. This usually depends on the specific retron system. For example, in case of certain retron systems such as retron Sen2, Eco4 and Eco7, the RNA is cleaved off, leaving behind just the ssDNA. In our case, with retron Eco2, the RNA remains stably bound to the ssDNA, thereby maintaining a stable hybrid RNA-DNA structure[10,13]. During the extraction of RT-DNA, the conjugated RNA is degraded during the RNase digestion step, and therefore is not visible in the gel images.

Reviewer #2 (Public Review):

Summary:

This manuscript explores a DNA fluorescent light-up aptamer (FLAP) with the specific goal of comparing activity in vitro to that in bacterial cells. In order to achieve expression in bacteria, the authors devise an expression strategy based on retrons and test four different constructs with the aptamer inserted at different points in the retron scaffold. They only observe binding for one scaffold in vitro, but achieve fluorescence enhancement for all four scaffolds in bacterial cells. These results demonstrate that aptamer performance can be very different in these two contexts.

Strengths:

Given the importance of FLAPs for use in cellular imaging and the fact that these are typically evolved in vitro, understanding the difference in performance between a buffer and a cellular environment is an important research question.

The return strategy utilized by the authors is thoughtful and well-described.

The observation that some aptamers fail to show binding in vitro but do show enhancement in cells is interesting and surprising.

We appreciate the reviewer's thorough assessment.

Weaknesses:

This study hints toward an interesting observation, but would benefit from greater depth to more fully understand this phenomenon. Particularly challenging is that FLAP performance is measured in vitro by affinity and in cells by enhancement, and these may not be directly proportional. For example, it may be that some constructs have much lower affinity but a greater enhancement and this is the explanation for the seemingly different performance.

We thank the reviewer for this insightful comment. In response, we conducted a series of additional control experiments to better understand the apparent discrepancy between the in vitro and in vivo data. These experiments revealed that the previously reported increase in intracellular green fluorescence is independent of retron-expressed Lettuce RT-DNA and DFHBI-1T, and instead reflects stress-induced autofluorescence of *E. coli* upon inducer and antibiotic treatment. Our original negative controls (empty wild-type *Eco2*, uninduced cells in the presence of DFHBI-1T) were therefore not sufficient to rule out this effect.

As a consequence, we have removed the earlier FACS data from the manuscript and no longer claim detectable intracellular Lettuce fluorescence. The reviewer's comment prompted us to re-examine the fluorogenicity of our constructs in vitro. We found that the 4Lev4 construct folds poorly and produces very low signal in in-gel staining assays with DFHBI-1T. In contrast, the 8LE variant (8-nt P1 stem at position v4) shows the highest fluorescence in these in-gel assays (new Figure 3C). Nevertheless, even this construct remains 100-fold less fluorogenic than the RNA-based FLAP Broccoli (new Figure 3-figure supplement 5), and we were unable to detect its intracellular fluorescence above background (new Figure 3-figure supplement 4).

To still directly demonstrate that retron-embedded Lettuce domains that are synthesized under intracellular conditions are functional, we modified our strategy in the revision and purified the expressed RT-DNA from *E. coli*, followed by in-gel staining with DFHBI-1T (new Figure 3E). Despite the challenge of obtaining sufficient amounts of ssDNA, this ex vivo approach clearly shows that the retron-produced Lettuce RT-DNA retains fluorogenic activity.

The authors only test enhancement at one concentration of fluorophore in cells (and this experimental detail is difficult to find and would be helpful to include in the figure legend). This limits the conclusions that can be drawn from the data and limits utility for other researchers aiming to use these constructs.

We appreciate this excellent suggestion. In the original experiments, the DFHBI-1T concentration in cells was chosen based on published conditions for live-cell imaging of the Broccoli RNA aptamer[14], which is substantially more fluorogenic than Lettuce. Motivated by the reviewer's comment, we explored different fluorophore concentrations and additional controls to optimize the in vivo readout. These experiments showed that the weak intracellular fluorescence signal is dominated by stress-induced autofluorescence[15] (possibly due to the weaker antitoxin activity of the modified *msd*) and does not depend on the presence of Lettuce RT-DNA or DFHBI-1T.

Given the combination of low Lettuce fluorogenicity and low intracellular RT-DNA levels, we concluded that varying the fluorophore concentration alone does not provide a meaningful way to deconvolute these confounding factors in cells. Instead, we shifted our focus to a more direct assessment of Lettuce activity: we now demonstrate that retron-produced Lettuce RT-DNA can be purified from *E. coli* and retains fluorogenic activity in an in-gel staining assay with DFHBI-1T (new Figure 3E). We believe this revised strategy provides a clearer and more quantitative characterization of the system's capabilities and limitations than the initial in vivo fluorescence measurements.

The FLAP that is used seems to have a relatively low fluorescence enhancement of only 2-3 fold in cells. It would be interesting to know if this is also the case in vitro. This is lower than typical FLAPs and it would be helpful for the authors to comment on what level of enhancement is needed for the FLAP to be of practical use for cellular imaging.

In the revised manuscript, we directly address this point by comparing the in vitro fluorescence of Lettuce (DNA) and Broccoli (RNA) under optimized buffer conditions. These experiments show that Broccoli is nearly two orders of magnitude more fluorogenic than Lettuce (new Figure 3-figure supplement 5). Thus, the low enhancement observed for Lettuce in cells is consistent with its intrinsically poor fluorogenicity in vitro.

Based on this comparison and on reported properties of RNA FLAPs such as Broccoli, we conclude that robust cellular imaging typically requires substantially higher fluorogenicity and dynamic range than currently provided by DNA-based Lettuce. In other words, under our conditions, Lettuce is close to or below the practical detection limit for in vivo imaging, whereas Broccoli performs well. We now explicitly state in the Discussion that further evolution and optimization of DNA FLAPs will be required to achieve fluorescence enhancements that are suitable for routine cellular imaging, and we position our work as a first demonstration that functional DNA aptamers can be produced in cells via retrons, while also delineating the current sensitivity limits.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Add gene accession numbers are not listed - how is this plasmid obtained?

The sequence was obtained from Millman et al[16], and ordered as gblock from IDT. The gblock was then cloned into a pET28a vector by Gibson assembly. We have now included this in the methods section.

Reviewer #2 (Recommendations For The Authors):

Page 2, line 40 - FLAPS should be FLAPs

We have corrected this typo in the revised version.

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